Mahad Mubashir Beg

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Python AI Engineer with 4+ years of professional software engineering experience and an MSc in Artificial Intelligence from the University of Surrey, building and shipping LLM powered products for clients. I build RAG systems with LangChain, FAISS and the OpenAI APIs, real time voice agents over WebRTC and VoIP with LiveKit, Twilio, Deepgram and ElevenLabs, and chatbots with NER on Groq LLaMA and fine-tuned GPT 4o. I also fine tune models with LoRA and PyTorch, and serve AI services through FastAPI on AWS with CI/CD pipelines. I work directly with clients to turn requirements into scalable and production ready AI solutions.

TECHNICAL SKILLS

AI/ML Frameworks | LangChain, HuggingFace Transformers, PyTorch, TensorFlow, FAISS, scikit-learn, UMAP, HDBSCAN, SHAP, LIME

Programming | Python, JavaScript/Node.js, TypeScript, C/C++, FastAPI, Flask, Django

Voice and Telephony | LiveKit, WebRTC, Twilio, FreeSWITCH, FusionPBX, Deepgram, ElevenLabs

Cloud & Deployment |AWS (Lambda, EC2, RDS, CodeBuild, EBS, S3), CI/CD, Docker, Kubernetes, RESTful API Integration, Gradio

Databases | Vector Databases (FAISS), PostgreSQL, MongoDB, MySQL, AWS RDS, Firebase, Redis, Kafka, RabbitMQ

AI Specializations | RAG Architectures, Prompt Engineering, Vector Search, Model Fine-Tuning (LoRA), Computer Vision, NLP, Explainable AI (XAI)

PROFESSIONAL EXPERIENCE

Senior Software Engineer (ML & Full Stack) | *Senarios PVT Ltd, Lahore, Pakistan* Sept 2023 – Sept 2025

● Built RAG-based multimodal voice bot platform where users create bots by uploading documents or website URLs; system scrapes content using BeautifulSoup, creates FAISS-based vector indexes, and enables real-time voice queries using Python FastAPI, LiveKit Agent, and OpenAI's low-latency API; integrated Twilio for number provisioning and call forwarding

● Developed accessible ride-booking chatbot using Python FastAPI with real-time Named Entity Recognition model using Groq LLaMA 3.1-70B-Versatile for dynamic address suggestions; employed few-shot prompt engineering and fine-tuned OpenAI's GPT-4o via LangChain for conversational interactions

● Led client meetings to gather requirements and delivered technical proposals providing tailored blockchain and AI solutions, contributing to product strategy and new client acquisition; facilitated SCRUM meetings fostering collaboration among cross-functional teams achieving timely delivery of project milestones

● Deployed scalable backend systems on AWS/EC2 with CI/CD pipelines; designed Polygon-based tokenized reward system using Solidity smart contracts for blockchain-powered fashion e-commerce platform integrating wallet solution

Tech: Pytho, FastAPI, Node.js, React, LangChain, OpenAI GPT-4o, Groq LLaMA, FAISS, BeautifulSoup, Twilio, LiveKit, AWS (EC2, RDS, CodeBuild), Solidity, Web3.js

Software Engineer | *Embedded Processing Limited, Lahore, Pakistan* Aug 2022 – Aug 2023

● Implemented WI-SUN communication protocol integration into Smart Meters' embedded C codebase, concurrently conducting rigorous unit testing through hardware flash programming.

● Automated testing utilizing VectorCAST to validate code integrity and reliability. | *Tech: C/C++, Vue.js, Embedded Systems*

PROJECTS

Poocho AI: Agentic AI Platform | *Senarios PVT Ltd* June 2025 – Sept 2026

● Built the backend for a unified AI powered call centre platform using Node.js, Express.js, PostgreSQL, MySQL and Redis, developing the REST APIs that supported IVR flows, live agent routing and automated campaign management.

● Implemented IVR flows and configured telephony using FreeSWITCH and FusionPBX, connecting VoIP call handling to the application backend. Used RabbitMQ for asynchronous call and event processing, and deployed services.

Morphological Toxicity Prediction | *Computer Vision* Sept 2025 – Sept 2026

● Designed a five-experiment ablation study on a biological dataset to benchmark supervised models (XGBoost, CatBoost) for multi-label toxicity classification & regression across 412 endpoints. Implemented compound-level stratified cross-validation with leakage prevention.

● Achieved state-of-the-art cytotoxicity prediction scores (MTT R^2=0.820, AUROC=0.619) using supervised mutual information feature selection (SelectKBest) recovering performance with 5.2× fewer features | *Python, scikit-learn, PCA, pandas, NumPy, HPC/SLURM*

Dialect-Aware Sarcasm Detection | *Natural Language Processing* Feb 2026 - May 2026

● LoRA fine-tuning of TinyLlama-1.1B (1.13M trainable params = 0.11% of base) for sarcasm detection across 3 English varieties; deployed Gradio web app with dynamic adapter switching (near-zero latency, no model reload)

● Achieved Macro-F1 0.6275 vs pooled RoBERTa 0.6075 with 550× smaller footprint. | *Tech: HuggingFace PEFT, LoRA, Gradio, PyTorch*

Diabetic Retinopathy Grading with Vision Transformers | *Applied Machine Learning* Feb 2026 - May 2026

● Designed ViT-B/16 (86M params) pipeline for 5-class medical image classification (35K retinal images); engineered two-phase fine-tuning with 7 systematic ablation experiments (FocalLoss, WeightedRandomSampler, Downsampling)
● Built explainability module (Grad-CAM, Attention Rollout, LIME) identifying clinical failure modes. | *Tech: PyTorch, timm, LIME*

PM2.5 Air Quality Forecasting System for Lahore, Pakistan | *AI & Sustainability* Feb 2026 – May 2026

● Designed end-to-end ML pipeline for air quality forecasting: data acquisition (NASA FIRMS MODIS API for 47,353 wildfire records), feature engineering (42 temporal/spatial features), model training (LSTM, CNN1D, Transformer, GRU), XAI analysis (SHAP, LIME), and edge deployment optimization
● Implemented TFLite quantization achieving 85-91% model compression while maintaining 90%+ accuracy for deployment on resource-constrained edge devices (mobile/IoT air quality monitors); best model (Transformer): QWK 0.501, RMSE 67.9 µg/m³
● Generated policy recommendations for government air quality management based on SHAP/LIME explainability analysis identifying wildfire proximity as key predictive feature. | *Tech: Python, PyTorch, LSTM, Transformer, SHAP, LIME, TFLite, NASA MODIS API*

Numbers AI: RAG based Multimodal Voice Bot Platform for Business Automation| *Senarios PVT Ltd* Jan 2025 - June 2025

● Built a customizable RAG voice bot platform where users can create bots by uploading documents (PDFs, Word files) or providing website URLs. The system scrapes content using BeautifulSoup and creates FAISS-based vector indexes to enable accurate, context-aware responses to real-time voice queries. Developed with Python FastAPI, LiveKit Agent, and OpenAI's low-latency API. Integrated Twilio for number provisioning and call forwarding. Supports multilingual voice interactions for lead generation, customer support, and campaign automation.

AI-Powered Restaurant Reservation and Order Management Voice Bot | *Senarios PVT Ltd* July 2024 – Dec 2024

● Engineered a voice bot using Deepgram for speech-to-text, Eleven Labs for text-to-speech, and OpenAI's real-time API for LLM-generated responses. Optimized model parameters to reduce latency and improve transcription accuracy, automating reservation booking and order-taking with scalable backend workflows.

IT Curves: Accessible Ride-Booking Chatbot with Advanced AI Integration | *Senarios PVT Ltd* Jan 2024 – June 2024

● Leveraged Python FastAPI framework to develop a chatbot facilitating ride bookings for customers with disabilities, enhancing accessibility and user experience. Implemented a real-time Named Entity Recognition model using Groq LLaMA 3.1-70B-Versatile for dynamic address suggestions. Employed few-shot prompt engineering techniques and fine-tuned OpenAI's GPT-4o model via LangChain for conversational interactions and booking management.

Video2Music: AI-Driven Video to Music Conversion Platform | *LUMS* Sept 2023 – Dec 2023

● Developed a full-stack web app using React and Python that allows users to upload videos, transcribe audio with WhisperAI. Fine-tuned GPT-3.5-T for mood analysis and synthesize AI-generated music using Meta's MusicGen.

Unveiling Road Accident Patterns and Trends: A Data Mining Approach | *LUMS* Feb 2022 – June 2022

● Performed EDA, statistical analysis and visualization, to understand accident trends. Applied clustering and frequent pattern mining algorithms to identify accident hotspots, developed data-driven recommendations for enhanced road safety.

A Neural Network's Exploration of Custom Image Data | *LUMS* Jan 2021 – June 2021

● The goal was to develop and evaluate a neural network for classifying fashion items from images. Built and evaluated a multi-layer perceptron, trained via backpropagation, on Fashion MNIST and a custom dataset. Applied normalization and one-hot encoding. Achieved high accuracy on fashion item recognition, reflecting expertise in neural network architecture, hyperparameter tuning, and model evaluation.

PUBLICATIONS & AWARDS

- **Graduated with Distinction**: MSc Artificial Intelligence
- **M. Mubashir,** Z. Afzal, and R. Rafay, "PoW to PoS Transition: Early Lessons from Ethereum's Merge", 2023.
- **Dean's Honour List:** Recognized for academic excellence in Spring 2020.
- **Football & Cricket Excellence:** Awarded the prestigious LUMS Football Team **Colour Award** and named **Top Goal Scorer** for two consecutive years. Awarded Top Runs Scorer for the LUMS Cricket Team.
- **A-Level Merit Scholarship:** Awarded a 100% merit-based scholarship for exceptional academic performance.
- **Academic Distinctions:** Internal Distinction in O-Level Chemistry and Straight A's in Cambridge O-Level exams.

EDUCATION

- **MSc Artificial Intelligence** - University of Surrey
- **BSc. Computer Science** - Lahore University of Management Sciences (LUMS)

Dissertation: *Diagnostic Characterisation of CP-CNN Feature Space Structure for Hepatotoxicity Prediction*

Coursework: NLP, Applied ML, Computer Vision and Pattern Recognition, AI & Sustainability, People-Centered AI, Data Structures, Algorithms, Operating Systems, Databases, Artificial Intelligence, Machine Learning Fundamentals, Data Science, Data Mining